MagicBox Playbook Reference

Reference for every Ansible recipe in this repository: purpose, variables, commands, tags, prerequisites, verification and operational risks. It matches the code at the time of the build; `tools/check_docs.py` fails if a role, variable or tag is missing from this document.

Pinned toolchain: ansible-core 2.21.5 (Python 3.14 `.venv` managed by uv), community.general 13.5.0, community.docker 5.4.0, community.crypto 3.5.1.

1. Overview

Item	Value
Playbooks	playbooks/site.yml (host + Portainer), playbooks/stacks.yml (default stacks), playbooks/update.yml (OS packages)
Inventory	inventory/hosts.yml (git-ignored; template inventory/hosts.example.yml)
Group	magicbox
Project policy	inventory/group_vars/magicbox/main.yml
Secrets	inventory/group_vars/magicbox/vault.yml (ansible-vault, git-ignored)
Collections	requirements.yml, installed to .ansible/collections
Result	Portainer CE at https://magicbox.local:9443/ with environment homelab; default stacks; LAN only

Play 1, *Configure MagicBox*, runs on group `magicbox` with `become: true` and applies the roles in this order: `preflight`, `mdns`, `firewall`, `docker`, `portainer`, `portainer_init`. Play 2, *Verify Portainer from the Ansible controller*, queries Portainer from the controller over the LAN.

`playbooks/stacks.yml` has one play, *Deploy default stacks through Portainer*, that runs the `stacks` role on group `magicbox`. Run it after `site.yml`.

`playbooks/update.yml` has one play, *Update MagicBox packages*, that runs the `os_update` role. It needs the vault password only for `vault_become_password`.

2. Commands

Goal	Command
Create the environment	make setup
All local checks	make check
Container converge + idempotency test	make test-container
First deployment	make deploy EXTRA="-e magicbox_confirm_destructive=true"
Routine deployment	make deploy
Only one area	uv run ansible-playbook playbooks/site.yml --ask-vault-pass --tags docker
Dry run with diff	uv run ansible-playbook playbooks/site.yml --ask-vault-pass --check --diff
Verification only	make verify
Default stacks	make stacks
Selected stacks only	uv run ansible-playbook playbooks/stacks.yml --ask-vault-pass -e '{"stacks_enabled": ["postgresql"]}'
Stacks: verification only	uv run ansible-playbook playbooks/stacks.yml --ask-vault-pass --tags verify
Stack status (read-only)	make status
Edit the vault	make vault-edit
OS package update	make update
OS update + reboot if required	make update EXTRA="-e os_update_reboot=true"
Idempotency on MagicBox	make idempotency VAULT_PASSWORD_FILE=<file>

Goal	Command
List tasks / tags	uv run ansible-playbook playbooks/site.yml --list-tasks / --list-tags

`make deploy`, `verify`, `stacks` and `update` ask for the vault password; the sudo password is `vault_become_password` (`ansible_become_password` in `inventory/hosts.yml`). `BECOME=-K` prompts for it instead.

A `--check` run cannot complete the `docker` and `portainer` roles on a fresh host because the packages they verify are not installed yet.

3. Tags

Tag	Selects
preflight	The preflight role
always	Also on preflight: it runs with any --tags selection, so checks and facts are always present
mdns	The mdns role
firewall	The firewall role
docker	The docker role
portainer	The portainer and portainer_init roles, including TLS tasks
portainer-init	Only the portainer_init role (site environment)
stacks	The stacks role (in playbooks/stacks.yml)
status	The read-only status report (in playbooks/status.yml)
update	The os_update role (in playbooks/update.yml)
tls	Only TLS provisioning inside portainer (certificate renewal, CA fetch)
verify	Verification tasks in all roles plus the controller play; in stacks.yml, login, lookups and stack checks
controller-verify	Only the controller play; skipped by the container test

`--tags verify` runs `preflight` (tagged `always`) and every `verify` task. It changes nothing on a converged host.

4. Shared variables

Defined in `inventory/hosts.yml` (environment facts) or `inventory/group_vars/magicbox/main.yml` (project policy).

Variable	Default	Description
magicbox_lan_interface	required	LAN interface on MagicBox (ip -br link). Avahi publishes only here.
magicbox_lan_cidr	required	Private IPv4 LAN subnet allowed to reach SSH, Portainer and mDNS.
ansible_ssh_extra_args	"-4"	SSH over IPv4: ufw allows only the IPv4 magicbox_lan_cidr; an IPv6 link-local connection is refused by preflight.
ansible_become_password	{{ vault_become_password }}	sudo password of ansible_user, set in inventory/hosts.yml; remove with passwordless sudo.
vault_become_password	required (vault)	sudo password; anyone with the vault password can become root on MagicBox.
ansible_become_exe	sudo	Ubuntu 26.04: sudo.ws (classic sudo; sudo-rs prompts time out, ADR-0002).

Variable	Default	Description
magicbox_hostname	magicbox	Hostname and mDNS name (<name>.local); single DNS label.
magicbox_expected_architecture	x86_64	Asserted CPU architecture.
magicbox_supported_os	Ubuntu 24.04/26.04, Debian 12/13	Asserted OS/version map (ADR-0002).
magicbox_confirm_destructive	false	Allows destructive steps for one run (ADR-0008). Pass with -e.
magicbox_site_name	homelab	Name of the Portainer environment (site) for the local Docker engine (ADR-0010).
magicbox_portainer_api_url	https://magicbox.local:9443	Portainer API used by portainer_init and stacks; keep in sync with portainer_https_port.
magicbox_portainer_ca_cert	/opt/magicbox/portainer/certs/ca.crt	CA used to validate the API; keep in sync with portainer_cert_dir.
portainer_admin_password	{{ vault_portainer_admin_password }}	Bootstrap admin password, from the vault.
vault_portainer_admin_password	required (vault)	At least 12 characters; in vault.yml only.
vault_postgresql_password, vault_rabbitmq_password, vault_grafana_admin_password	required for the matching stack (vault)	At least 12 characters each.

5. Role `preflight`

Purpose. Fail fast on unsupported or misconfigured hosts and prepare the baseline: OS/architecture/inventory/vault assertions, LAN CIDR validation, detection of where the SSH controller connects from, hostname, base packages.

Prerequisites. `magicbox.local` resolves from the controller (Avahi installed during the manual steps); SSH key login as a sudo-capable user; Python 3 on the host (present on Ubuntu/Debian server installs).

Variable	Default	Description
preflight_base_packages	ca-certificates, curl, gnupg, iproute2, python3-cryptography, python3-debian, python3-requests	Packages needed by later roles and modules.
preflight_manage_hostname	true	Set the static hostname and 127.0.1.1 entry. false only in the test container.

Facts set. `preflight_controller_location`: `inside`, `outside` or `unknown` (no `SSH_CLIENT`, e.g. the container test), consumed by `firewall`.

Verification. All assertions run on every play (tag `always`).

Risks. Changing `magicbox_hostname` renames the host and the mDNS name; certificates must match (`portainer_tls_dns_names` follows automatically, the existing CA's name constraints do not: rotate the CA, installation guide 8.6).

6. Role `mdns`

Purpose. Publish `magicbox.local` on the LAN with Avahi (ADR-0003). MagicBox has no fixed IP: Avahi re-announces the name whenever DHCP changes the address. Avahi is first installed by hand (installation guide 1.5) so Ansible can connect by name; this role then owns its configuration. Installs `mdns_packages`, disables systemd-resolved's mDNS/LLMNR responders when systemd-resolved is installed, templates `/etc/avahi/avahi-daemon.conf` (host name, `allow-interfaces`, no HINFO/workstation publishing, no reflector, no wide-area), and enables `avahi-daemon`.

Variable	Default	Description
mdns_use_ipv6	false	Publish AAAA records. Off because the firewall only allows the IPv4 LAN.
mdns_packages	avahi-daemon, avahi-utils, libnss-mdns	Responder, CLI tools, NSS module for local resolution.

Handlers. *Restart systemd-resolved*, *Restart avahi-daemon*.

Verification. `avahi-resolve-host-name -4 magicbox.local` must return an address (retries 5 x 2 s).

Risks. Avahi renames the host to `magicbox-2.local` if another device already claims the name; TLS then fails the name check. Restricting to `magicbox_lan_interface` means the name is invisible on other interfaces.

7. Role `firewall`

Purpose. Enable ufw with default deny inbound and allow only SSH, `firewall_lan_tcp_ports` and `firewall_lan_udp_ports` from `magicbox_lan_cidr` (ADR-0004). Rules persist across reboots.

Variable	Default	Description
firewall_ssh_port	22	SSH port allowed from the LAN.
firewall_lan_tcp_ports	[9443]	TCP ports allowed from the LAN (Portainer).
firewall_lan_udp_ports	[5353]	UDP ports allowed from the LAN (mDNS).
firewall_logging	low	ufw logging level.
firewall_docker_user_filter	true	Manage the DOCKER-USER block in /etc/ufw/after.rules that limits Docker-published ports to the LAN (ADR-0012).

Safety rails. Refuses first activation unless `magicbox_confirm_destructive=true`; always refuses when `preflight_controller_location` is `outside`.

Handlers. *Reload ufw* when the DOCKER-USER block changes.

Verification. `iptables -S DOCKER-USER` contains the LAN RETURN rule and the interface DROP rule. `sudo ufw status verbose` (manual); the controller play proves 9443 is reachable from the LAN.

Risks. A wrong `magicbox_lan_cidr` can lock out SSH from other subnets (console recovery: installation guide 8.3). Rules for ports no longer listed are not removed automatically; delete them with `sudo ufw delete <n>`. Ports published by containers bypass ufw's INPUT rules; the DOCKER-USER block restricts them to `magicbox_lan_cidr` instead. Setting `firewall_docker_user_filter: false` exposes them to any source that can route to MagicBox.

8. Role `docker`

Purpose. Install Docker Engine from Docker's apt repository with a verified signing key and a major-version pin, configure the daemon, and enable `containerd` and `docker` at boot (ADR-0002).

Variable	Default	Description
docker_major_version	"29"	apt preference pin 5:29.* for docker-ce, cli, rootless-extras.
docker_apt_key_url	https://download.docker.com/linux/<distro>/gpg	Signing key URL.
docker_apt_key_fingerprint	9DC858229FC7DD38854AE2D88D81803C0EBFCD88	Expected key fingerprint; mismatch aborts.
docker_apt_key_path	/etc/apt/keyrings/docker.asc	Keyring path referenced by the deb822 source.

Variable	Default	Description
docker_packages	docker-ce, docker-ce-cli, containerd.io, docker-buildx-plugin, docker-compose-plugin	Installed packages.
docker_conflicting_packages	docker.io, docker-doc, docker-compose, docker-compose-v2, podman-docker, containerd, runc	Removed only with confirmation.
docker_log_max_size	10m	json-file log rotation size.
docker_log_max_file	3	json-file rotated files kept.

`daemon.json` also sets the log option `tag` to the container name, which Alloy uses as the `container` label (ADR-0013).

Files. `/etc/apt/sources.list.d/docker.sources`, `/etc/apt/preferences.d/docker-ce.pref`, `/etc/docker/daemon.json` (validated with `dockerd --validate` before replacement; `live-restore` enabled).

Handlers. *Restart docker* (containers survive thanks to live-restore).

Verification. No `dockerd` TCP listener (`ss -Hltnp`).

Risks. Removing conflicting packages stops workloads that used them. A daemon restart briefly interrupts the Docker API. A major upgrade requires changing `docker_major_version` deliberately.

9. Role `portainer`

Purpose. Run Portainer CE with persistent data, HTTPS on 9443 using the MagicBox CA, plain HTTP disabled, and a pre-set admin password (ADR-0004, ADR-0005, ADR-0006). Container: `portainer`, image `portainer/portainer-ce:2.45.3`, restart policy `unless-stopped`, `network_mode: host`; mounts the Docker socket, the `portainer_data` volume, the certificate directory (read-only) and the password file (read-only).

Variable	Default	Description
portainer_image	portainer/portainer-ce	Image repository.
portainer_version	"2.45.3"	Image tag (LTS); also asserted against the API.
portainer_container_name	portainer	Container name.
portainer_https_port	9443	HTTPS port (keep in firewall_lan_tcp_ports).
portainer_data_volume	portainer_data	Persistent named volume mounted at /data.
portainer_base_dir	/opt/magicbox/portainer	Host base directory.
portainer_cert_dir	<base>/certs	TLS material (0700).
portainer_secret_dir	<base>/secrets	Admin password file (0700).
portainer_tls_mode	local_ca	local_ca or provided.
portainer_tls_dns_names	magicbox.local, magicbox	Certificate SANs and CA name constraints.
portainer_tls_ip_addresses	[]	Optional IP SANs/constraints. Keep empty: the IP is dynamic (DHCP).
portainer_tls_ca_common_name	MagicBox Local CA (magicbox)	CA subject CN.
portainer_tls_ca_valid_for	+3650d	CA lifetime.
portainer_tls_cert_valid_for	+397d	Server certificate lifetime.
portainer_tls_renew_before	+30d	Re-issue window before expiry.
portainer_tls_ca_fetch_dest	artifacts/magicbox-local-ca.crt	Controller copy of the CA for clients.
portainer_tls_provided_cert_src	empty	provided mode: server certificate path on the controller.
portainer_tls_provided_key_src	empty	provided mode: server key path (keep it vaulted).

Variable	Default	Description
portainer_tls_provided_ca_src	empty	provided mode: issuing CA path.

TLS tasks (`tls_local_ca.yml`). CA key (ECDSA P-384) and self-signed CA (`CA:TRUE, pathlen:0`, critical name constraints) created once; server key (ECDSA P-256), CSR and certificate (EKU `serverAuth`) signed by the CA and re-issued when within `portainer_tls_renew_before` of expiry or when the SANs change; CA fetched to the controller.

Handlers. *Restart portainer* when TLS material changes.

Verification. `GET https://magicbox.local:9443/api/system/status` from MagicBox with CA validation (retries 20 x 3 s) and an assertion that the reported version equals `portainer_version`. The controller play repeats the request over the LAN.

Risks. Portainer controls the Docker socket and is root-equivalent: every Portainer admin is root on MagicBox. Changing the vault password after first start does not change the admin password. Deleting `portainer_cert_dir` creates a new CA that clients must trust again. Deleting the `portainer_data` volume erases all Portainer settings; the playbook never does this.

10. Role `portainer_init`

Purpose. Initialise Portainer through its API: log in as admin with the vault password and ensure one local Docker environment named `magicbox_site_name` (`homelab`). It renames an existing local-socket environment, or creates one (ADR-0010). `tasks/api_login.yml` is reused by `stacks`.

Variable	Default	Description
portainer_init_api_user	admin	Portainer user for API calls; password is portainer_admin_password.

Prerequisites. The `portainer` role has run (Portainer up, CA present).

Verification. The environment exists and its URL is `unix:///var/run/docker.sock`.

Risks. Fails at login if the admin password was changed in the UI and not in the vault. Changing `magicbox_site_name` after stacks exist creates a new, empty environment; rename the old one in the UI first.

11. Role `stacks` (playbooks/stacks.yml)

Purpose. Deploy the default services as Portainer standalone stacks in the site environment (ADR-0011). For each name in `stacks_enabled` it renders `roles/stacks/templates/<name>.yml.j2`, passes the secrets as stack environment variables, and creates the stack, or updates it when the compose file or environment differs or one of its containers is missing (e.g. an interrupted first deployment). Config files for `observability` are written to `stacks_base_dir`; their checksum (`MAGICBOX_CONFIG_REVISION` label) forces a redeploy when they change.

Stack	Containers	Published ports	Volumes
rabbitmq	rabbitmq	5672, 15672	rabbitmq_data
postgresql	postgresql	5432	postgresql_data
ollama	ollama	11434	ollama_data
observability	grafana, loki, alloy	3000 (Grafana HTTPS)	grafana_data, loki_data, alloy_data

Variable	Default	Description
stacks_enabled	rabbitmq, postgresql, ollama, observability	Stacks to manage. Removing one does not delete it.
stacks_base_dir	/opt/magicbox/stacks	Host directory for mounted config files.
stacks_rabbitmq_image	rabbitmq:4.3.7-management	RabbitMQ image.
stacks_postgresql_image	postgres:18.6	PostgreSQL image (newest major, supported to 2030).
stacks_ollama_image	ollama/ollama:0.40.12	Ollama image.
stacks_grafana_image	grafana/grafana:13.2.5	Grafana image.
stacks_loki_image	grafana/loki:3.7.9	Loki image.

Variable	Default	Description
stacks_alloy_image	grafana/alloy:v1.21.2	Alloy image (Promtail replacement, ADR-0013).
stacks_rabbitmq_amqp_port	5672	Host port for AMQP.
stacks_rabbitmq_management_port	15672	Host port for the RabbitMQ UI/API (HTTP).
stacks_postgresql_port	5432	Host port for PostgreSQL.
stacks_ollama_port	11434	Host port for the Ollama API.
stacks_grafana_port	3000	Host port for Grafana (HTTPS).
stacks_rabbitmq_user	magicbox	RabbitMQ default user.
stacks_rabbitmq_password	{{ vault_rabbitmq_password }}	RabbitMQ password (first start only).
stacks_postgresql_user	magicbox	PostgreSQL superuser.
stacks_postgresql_database	magicbox	Database created at first start.
stacks_postgresql_password	{{ vault_postgresql_password }}	PostgreSQL password (first start only).
stacks_grafana_admin_password	{{ vault_grafana_admin_password }}	Grafana admin password (first start only).
stacks_ollama_models	[]	Models to pull after deployment, e.g. ["llama3.2:3b"]; never deleted.
stacks_loki_retention	168h	Loki log retention.

Internal data (`roles/stacks/vars/main.yml`, not meant to be overridden): `stacks_env` (environment per stack), `stacks_required_secrets`, `stacks_containers`.

Prerequisites. `playbooks/site.yml` completed (Portainer, `homelab` environment, DOCKER-USER filter); the vault contains the stack passwords.

Verification. All stack containers running, and healthy where a healthcheck exists (up to 15 minutes, to cover first image pulls). PostgreSQL `pg_isready`. RabbitMQ `/api/overview` with the vault credentials. Ollama `/api/version`. Grafana `/api/health` over HTTPS with the MagicBox CA. Loki has a `container` label value `loki`, which proves Alloy -> Loki -> Grafana works.

Status. `make status` (`playbooks/status.yml`, `tasks/status.yml`) changes nothing: it lists each image of the enabled stacks as OK or MISSING and each container as OK, NOT OK (stopped, restarting, unhealthy) or MISSING, and fails when anything is not OK.

Risks.

- Passwords are applied only on first initialisation of each data volume. Changing a vault value later breaks login and verification until the service password is changed too.
- Stack environments (including passwords) are stored in Portainer's database.
- Edits made to these stacks in the Portainer UI are overwritten by the next run.
- RabbitMQ, PostgreSQL and Ollama use plain-text protocols on the LAN; Ollama has no authentication. Access is limited to the LAN by DOCKER-USER.
- Grafana uses a copy of the Portainer certificate. After a certificate renewal by `site.yml`, run `stacks.yml` to refresh the copy.
- First deployment downloads about 4 GB (Ollama is about 3.4 GB).

12. Role `os_update` (playbooks/update.yml)

Purpose. Keep the OS current: `apt update`, `apt upgrade` and `apt autoremove`. Docker CE updates stay within `docker_major_version` (apt pin); `live-restore` keeps containers running while dockerd restarts.

Variable	Default	Description
os_update_upgrade	safe	safe (apt upgrade, never removes packages) or full (apt full-upgrade).
os_update_autoremove	true	Remove packages that are no longer needed.

Variable	Default	Description
os_update_reboot	false	Reboot when /var/run/reboot-required exists.
os_update_reboot_timeout	900	Seconds to wait for MagicBox to return after the reboot.
os_update_services	avahi-daemon, docker	Must be running after the update; units not installed yet are skipped.

Verification. Every installed unit in `os_update_services` is running after the update (and after the reboot, if any). Before the first `make deploy`, Docker is not installed and is skipped.

Risks.

- A reboot stops Portainer and every stack for a few minutes; they restart automatically. Hence `os_update_reboot` is off by default and the play only reports a pending reboot.
- `full` may remove packages; review `apt list --upgradable` on MagicBox first.

13. Testing the recipes

Check	Command	Proves
Syntax	make syntax	Playbook parses with the example inventory
Lint	make lint	yamllint + ansible-lint production profile
Secrets	make secrets-check	No plaintext vault values or private keys tracked
Docs	make docs-check	Links, ADR registry, this reference covers all roles/variables/tags
Container	make test-container	Full converge and changed=0 second run in an Ubuntu 24.04 systemd container (arm64 or amd64 per workstation)
Container + stacks	make test-container ARGS="--with-stacks"	Also runs stacks.yml twice (deploy + changed=0)
Host	make deploy, make idempotency, make verify	Real behaviour on MagicBox

The container test overrides `magicbox_expected_architecture`, `magicbox_lan_interface`, `magicbox_lan_cidr` and sets `preflight_manage_hostname: false`; it cannot prove LAN mDNS or reachability from other machines.

14. Changing the recipes

1. Edit the role; keep variables prefixed and documented in `defaults/main.yml`.
2. Update this reference (roles, variables, tags, risks) and, for significant decisions, add an ADR and list it in the registry and `CLAUDE.md`.
3. `make manuals check test-container`.
4. Commit sources and rebuilt PDFs together.